Compartmentalising the tank can significantly reduce the effect of slosh by allowing the compartments to be unloaded one at a time. Only a proportion of the load is free to slosh at any one time and so the effect is reduced. It has been shown (UMTRI, 2001) that if no more than 20% of the total load is free to slosh at any one time the rollover stability of the partially loaded vehicle will always be greater than that of a fully loaded vehicle. Current New Zealand regulations for hazardous goods transport require compartments of less than 10,000 litres if fitted with baffles (these are transverse baffles) or less than 7,000 litres if not fitted with baffles. In practice this means that typical petrol tanker semi- trailers have more than five compartments and thus, provided they are unloaded correctly, will always have better rollover stability when partially laden than when fully laden. This is not necessarily the case for tankers not carrying hazardous goods. To facilitate cleaning, milk tankers are generally not compartmentalised. They are usually fitted with baffles but only in the transverse direction. In their favour most of their travel distance is done with the tank either empty or fully laden. Hanging Meat Loads In hanging meat loads, the meat is suspended by hooks on rails suspended from the ceiling or walls of a refrigerated van body. In New Zealand there are two basic types of vehicle configuration used for transporting hanging meat. The first of these is a dedicated vehicle in which case the rails run longitudinally along the length of the vehicle with, typically, five rails across the width of the vehicle. These vehicles are typically a little over 2m from floor to ceiling because the driver has to be able to reach the rails to take carcasses on and off. Two of the vehicle/loads we inspected were of this type (see Figure 4 and Figure 5). In both these cases the floor-to-ceiling height was 2.1m. The alternative is a general- purpose vehicle in which case removable cross-beams are fitted with slots that act as transverse rails. A single longitudinal rail is used as a feeder to the transverse rails. The third load we inspected was of this type as shown in Figure 6. The feeder rail runs along the left hand side of the vehicle. General-purpose vehicles typically have a greater internal height to allow for other high volume loads but the hanging meat rails are hung at a height where the driver can reach them and so there is relatively little difference in payload height between these vehicles when carrying a hanging meat load. In the particular example shown in Figure 6 the transverse beams are 2.25m above the floor. Broadly speaking, the transport of hanging meat can be categorised as involving one of two freight tasks; the line haul task where meat is moved from a meat processing facility to a cool store or distribution centre and the local delivery task where meat is moved from a distribution centre to retail outlets such as supermarkets or butchers' shops. The line haul task generally involves full vehicle loads of a single species while local delivery vehicles may carry a mix of species and may make multiple stops so that, at least some of the time, they are only partially loaded. In practice this distinction is not so clear cut. Line haul loads may involve multiple drop-off points so that significant parts of the journey are done with part-loads. Local delivery operations are not limited to urban driving and may involve open-road speeds with high-speed cornering so rollover stability is important for these loads as well. Figure 4 shows a line haul load of pig carcasses which was transported from the South Island to Auckland. In the case parts of the load were delivered to several North Island centre en-route so that the load as photographed is only a part-load. Figure 5 shows a typical local delivery hanging meat load showing the less tight packing and the mixture of species, while Figure 6 is also a line haul part load.